



VERITAS UNIVERSITY, ABUJA
DEPARTMENT OF SOFTWARE ENGINEERING
FIRST SEMESTER 2025/2026 SESSION
IFT 211: DIGITAL LOGIC DESIGN - THEORY

Answer Question 1 and any other 2

16/02/2026

Time Allowed: 2 hours

Question 1. ✓

- (i) Convert 11011.101_2 to decimal (ii) Convert 4057.06_8 to decimal. (iii) Convert 378.93_{10} to octal.
- (iv) Convert 105.15_{10} to binary (v) Convert 756.603_8 to hexadecimal.
- (vi) Add the binary numbers 1101.101 and 111011 . (vii) Subtract 111.111_2 from 1010.01_2 .
- (viii) Multiply 1011.101_2 by 101.01_2 (ix) Divide 110101.11_2 by 101_2 (x) Convert $(1001)_{BCD}$ to Excess-3.
- (xi) Mention the ten different types of binary codes. (xii) Define decoder and Name 4 examples.
- (xiii) List the advantages and disadvantages of BCD code.
- (xiv) Mention the advantages of ASCII code. (xv) List the five characteristics of digital Ics.

Question 2. ✓

- (i) Define and Prove the Universality Property of NAND and NOR gates. (Show the Truth Table, Boolean Expressions and Draw the Logic Gates) to obtain:
- (ii) NAND implementation of NOT function (iii) NAND implementation of AND function (iv) NAND implementation of OR function (v) NOR implementation of NOT function (vi) NOR implementation of AND function (vi) NOR implementation of OR function.

Question 3. ✓

- (i) Explain Sum Of Product (SOP) and Product Of Sum (POS) forms using examples.
- (ii) State and prove the two De Morgan's Theorems showing the boolean expression, logic circuit and Truth Table.
- (iii) Simplify the following using DeMorgan's theorems:

(a) $(A+B)(\overline{A+B})$ (b) $\overline{\overline{A} \overline{B} \overline{C} \overline{D}}$

- (iv) Write the Equation form, Logic Function Table and draw K-Map showing the minterms for:
 (a) 2- variable K-Map (b) 3-Variable K-Map (c) 4-variable K-Map.

Question 4.

- (i) Minimize the function $F(w,x,y,z)$ using algebraic modifications. Show the result as a sum of products with a minimum number of literals. $\{F(w,x,y,z) = (y'+wy'(xz+xz'))z + (xy'+(x+y)'z')' + y(z+wx'(w'+z))\}$
- (ii) Convert the original function $F(w,x,y,z)$ from part (i) to a sum-of-product form and minimize it using Karnaugh map.
- (iii) Minimize the following expression using K-map:
 $Y = \overline{A} \overline{B} \overline{C} D + \overline{A} B \overline{C} D + \overline{A} B C D + \overline{A} B C \overline{D} + A B \overline{C} \overline{D} + A B \overline{C} D + A B C D + A \overline{B} C D.$
- (iv) $AB'C + A'BC + A'B'C + A'B'C' + AB'C'$

Question 5

- (i) Define Combinational Circuit with diagram and explain the 3 characteristics.
- (ii) Explain the Full Adder showing the Truth Table and Logic Gate circuit and the output algebraic function.
- (iii) Briefly describe a Multiplexer and give the Truth Table, Logical Expression and Logic Circuit of a 4X1 multiplexer.
- (iv) List the 6 different types of flip-flops.